

# Critical Summary: Decentralized Detection with Many Sensors

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# 1 Introduction

## 1.1 Background - Remote Hypothesis Testing

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## 1.2 2

In our work, we are going to discuss a problem of Decentrized Hypothesis Testing, where many sensors observe an environment, compress the observation into an action, and send the actions to a fusion center (FC). The FC decides between two hypothesis -  $H_1$  or  $H_2$ .

The paper we are going to discuss is Decentrized Detection with Many Sensors: Optimality of Exchangeable and Identical Encoding Policies [1], and specifically, we are attending the finite number of sensors problem discussed in the paper.

## 2 Problem Setup

We consider a problem of Binary Detection, as described in [1]. Assuming there are two hypothesis,  $H_1$  and  $H_2$ , the true one can be regarded as  $H^*$ , a random variable with  $P(H^* = H_1) = p$ ,  $P(H^* = H_2) = 1 - p$ . The setup being tested:

- $N < \infty$  sensors, each makes an observation of the environment denoted by  $y^i \in \mathcal{Y}^i$  as the observation of the  $i$ -th sensor for  $i \in \{1, \dots, N\}$ .
- Each sensor quantizes its observation using a mapping  $\gamma^i : \mathcal{Y}^i \rightarrow \mathcal{U}^i$ ,  $\mathcal{U}^i$  being the set of actions the sensor can send.  $\gamma^i$  is the decision rule / policy of the  $i$ -th sensor.
- The FC's policy is noted as  $\gamma^0 : \mathcal{U}^0 \times \mathcal{U}^1 \times \dots \times \mathcal{U}^N \rightarrow \{1, 2\}$ . Essentially, the fusion center decides  $u_0 = \gamma^0(u^{1:N})$ ,  $u_0 \in \{1, 2\}$ .
- The cost function we will focus on is:

$$c(H_j, u^0) = \begin{cases} 1 & \text{if } u^0 \neq j \\ 0 & \text{if } u^0 = j \end{cases} \quad (1)$$

- Our goal is to minimize the expected cost:

$$J^N(\gamma^{0:N}) \triangleq \mathbb{E}[c(H, u^0)] \quad (2)$$

As the number of sensors  $N$  increases, according to [1] and [2],  $J^N(\gamma^{0:N})$  is bound by a decaying exponent, regardless of the policies being used. This decay, and precisely how strong this decay is, is what carries the relevant information about the performance of a vector of policies, so the problem of minimizing the expected cost becomes maximizing the normalized error exponent -

$$J_{EE}^N \left( \frac{\log(J^N(\gamma^{0:N}))}{N} \right) \quad (3)$$

## 2.1 Assumptions

In order to make the problem tractable, the authors of the paper consider a few assumptions:

- (i)  $y^1, \dots, y^N$  are i.i.d conditioned on  $H$
- (ii) Under either hypothesis, sensor  $i$ 's observation has a probability density, measured against one common reference measure shared across both hypothesis.
- (iii)  $\mathcal{U}^i = \mathcal{U}$  for  $i = 1, \dots, N$ , and  $\mathcal{U}$  is a finite set.
- (iv) The likelihood ratio:

$$l^i = L(y^i) := \frac{f_{y^i|H}(y^i|H_2)}{f_{y^i|H}(y^i|H_1)} \text{a.s.} \quad (4)$$

Is well behaving and invertible.

While (i) is a common assumption in order to simplify the math and making it tractable, assumptions (ii) and (iv) are essential in order for  $l^i$  to be a sufficient statistic of  $y^i$ , as will be discussed later. Assumption (iii) dictates that all sensors have the same action set, underlying assumption about the compression and rate being used. This will be discussed furthermore in section \*, where we will offer some modifications for the solution setup.

## 3 Sufficient Statistics, Existence of an Optimal Encoding Policy

### 3.1 Exponential Decay of the Error Probability

### 3.2 $l^i$ is a Sufficient Statistic

$l^i$  is a sufficient statistic of  $H$ : once its value is known, nothing else about the raw observation  $y^i$  can further move one's belief about which hypothesis is true. With only two hypotheses on the table, the entire way in which the observation's density can vary with  $H$  is exhausted by a single comparison - how large the density is under  $H_2$  relative to how large it is under  $H_1$ , at that particular observation. A single number is therefore inherently enough to hold everything that changes with  $H$ , and that number is, by construction, exactly  $l^i$ . (How this shows up concretely inside the paper's own machinery - via Lemma 1 - is worth treating separately once we get there.)

### No information loss: a Markov-chain and DPI reading

Sufficiency means that any two raw observations producing the same  $l^i$  are, for the purpose of distinguishing  $H_1$  from  $H_2$ , interchangeable: whatever detail of  $y^i$  is discarded in forming  $l^i$  carries no evidential weight about  $H$ . This has a clean information-theoretic statement. Since  $l^i$  is a deterministic function of  $y^i$ , the chain

$$H \rightarrow y^i \rightarrow l^i$$

is trivially Markov, and the Data Processing Inequality (DPI) gives

$$I(H; l^i) \leq I(H; y^i). \quad (5)$$

Equality holds iff  $l^i$  is sufficient for  $H$  - equivalently, iff  $H \rightarrow l^i \rightarrow y^i$  is *also* a Markov chain, i.e.  $y^i$  carries no further information about  $H$  once  $l^i$  is fixed, gives equality

$$I(H; l^i) = I(H; y^i). \quad (6)$$

### 3.3 Optimality of Threshold Policies and the MAP Decoder

#### Lemma 1

Let Assumption 1 hold. Then, for any fusion rule  $\gamma^0 \in \Gamma^0$ ,

$$\inf_{\gamma^{1:N} \in \Gamma^N} J^N(\gamma^0, \gamma^{1:N}) = \inf_{\gamma^{1:N} \in \Gamma_{TS}^N} J^N(\gamma^0, \gamma^{1:N}). \quad (7)$$

Where  $\Gamma_{TS}^N$  is the class of threshold policies. Fixing every sensor's policy except sensor  $i$ 's, sensor  $i$ 's optimization is:

$$\inf_{u^i \in \mathcal{U}} \left\{ \sum_{j=1,2} P(H_j | y^i) E_{\gamma_{-i}^*} [c(H_j, \gamma^0(u^{1:N})) | H_j] \right\} \quad (8)$$

By Bayes' rule,  $P(H_j | y^i)$  depends on  $y^i$  only through the likelihood ratio  $l^i$  – this is exactly the sufficiency of  $l^i$  from Section [X] showing up again, now inside the optimization itself. Substituting reduces the problem to

$$\inf_{u^i \in \mathcal{U}} \left[ g^i(H_1, u^i) + g^i(H_2, u^i) l^i \frac{P(H_2)}{P(H_1)} \right], \quad (9)$$

where  $g^i(H_j, u^i)$  is the expected cost of action  $u^i$  under  $H_j$  given the other sensors' fixed policies. This is linear in  $l^i$  for each action, so the best action is whichever line is lowest at that value of  $l^i$ , switching only at the points where lines cross – exactly a threshold rule.

#### Lemma 2

### 3.4 Existence of an Optimal Solution

Lemmas 1 and 2 tell us where to look; Theorem 1, via the compactness result of Lemma 3, is what guarantees something is actually there to find

## 4 Considering Rate